

Cybersecurity Career Roadmap

A complete, self-paced path from beginner to job-ready — using 100% free resources

How to use this roadmap

This roadmap is organized into 6 phases. Each phase lists what to learn, hands-on practice to do, and free resources to learn from. Go in order — don't skip the fundamentals. Spend real time practicing in labs, not just watching videos. Expect 8–14 months of consistent part-time study (5–10 hrs/week) to reach a job-ready entry-level skill set.

Phase 1 — Foundations (Weeks 1–6)

Goal: Build the IT fundamentals every security role assumes you already know.

Topics to cover

- How computers work: CPU, memory, storage, OS basics (Windows & Linux)
- Networking fundamentals: OSI model, TCP/IP, subnetting, DNS, DHCP, HTTP/HTTPS
- Linux command line: file system, permissions, processes, package managers, bash scripting basics
- Basic programming/scripting: Python fundamentals (variables, loops, functions, file I/O)

Free resources

Networking & IT Fundamentals

- Professor Messer — CompTIA Network+ free video course ([professormesser.com](https://www.professormesser.com))
- Cisco Networking Academy — free 'Networking Basics' course ([netacad.com](https://www.netacad.com))
- freeCodeCamp YouTube — 'Computer Networking Full Course'

Linux

- Linux Journey (linuxjourney.com) — free interactive Linux fundamentals
- OverTheWire: Bandit wargame (overthewire.org/wargames/bandit) — learn Linux by hacking challenges

Python for Security

- freeCodeCamp — 'Python for Everybody' full course (YouTube)
- Automate the Boring Stuff with Python — free online book (automatetheboringstuff.com)

Phase 2 — Core Security Concepts (Weeks 7–12)

Goal: Understand the CIA triad, common attacks, and security principles employers expect.

Topics to cover

- CIA Triad, risk management, defense-in-depth, least privilege
- Authentication vs authorization, encryption basics (symmetric/asymmetric, hashing)
- Common attack types: phishing, malware, social engineering, MITM, DoS
- Security frameworks: NIST CSF, CIS Controls, MITRE ATT&CK; (overview level)

Free resources

Structured Courses

- Professor Messer — CompTIA Security+ full free video course ([professormesser.com](https://prolessor.com))
- IBM 'Introduction to Cybersecurity Tools & Cyber Attacks' — free audit on Coursera
- Google Cybersecurity Certificate — free audit track on Coursera (google.com/cybersecurity)

Reference Frameworks

- MITRE ATT&CK; framework — attack.mitre.org (free, industry-standard)
- NIST Cybersecurity Framework — nist.gov/cyberframework (free PDF)

Phase 3 — Hands-On Practice & Tools (Weeks 13–20)

Goal: Get real hands-on experience with the tools security professionals use daily.

Topics to cover

- Setting up a home lab (VirtualBox/VMware + Kali Linux + vulnerable VMs)
- Network scanning & enumeration: Nmap, Wireshark
- Vulnerability scanning: OpenVAS, Nessus Essentials (free tier)
- Basic penetration testing concepts: reconnaissance, exploitation, reporting
- SIEM basics: log analysis, Splunk Fundamentals (free)

Free resources

Hands-On Labs

- TryHackMe — free tier with 'Pre Security' & 'Cyber Security 101' learning paths (tryhackme.com)
- Hack The Box Academy — free tier modules (academy.hackthebox.com)
- PortSwigger Web Security Academy — completely free, industry-respected (portswigger.net/web-security)
- VulnHub — free downloadable vulnerable VMs to practice on (vulnhub.com)

Tool-Specific Learning

- Wireshark University — free tutorials ([wireshark.org](https://www.wireshark.org))
- Splunk Fundamentals 1 — free course (education.splunk.com free tier)
- Nmap official guide — free (nmap.org/book)

Phase 4 — Specialization Track (Weeks 21–30)

Goal: Pick a direction. Cybersecurity is broad — choose 1 track to go deep in first.

Choose one specialization

- **Offensive Security (Pentesting/Red Team):** Web app hacking, exploitation, privilege escalation
- **Defensive Security (Blue Team/SOC Analyst):** Log analysis, incident response, threat detection
- **GRC (Governance, Risk & Compliance):** Risk assessments, audits, compliance frameworks
- **Cloud Security:** Securing AWS/Azure/GCP environments (pairs well with the Cloud Computing roadmap)

Free resources by track

Offensive Security

- TryHackMe 'Offensive Pentesting' path (free tier modules)
- PentesterLab free exercises (pentesterlab.com)
- OWASP Top 10 — free official documentation (owasp.org)

Defensive Security / SOC

- LetsDefend.io — free tier SOC analyst simulations (letsdefend.io)
- TryHackMe 'SOC Level 1' path (free modules)
- Blue Team Labs Online — free tier (blueteamlabs.online)

GRC

- ISC2 Certified in Cybersecurity — free training + free exam (isc2.org/certified-in-cybersecurity)
- NIST 800-53 & ISO 27001 overviews — free official summaries

Phase 5 — Free Certifications & Proof of Skill (Weeks 31–36)

Certifications validate your knowledge. These options are free or near-free:

Free / Low-Cost Certifications

- ISC2 Certified in Cybersecurity (CC) — 100% free exam, great entry credential
- Google Cybersecurity Certificate — free with Coursera financial aid audit
- TryHackMe & HTB Academy completion badges — free, showcase on LinkedIn
- Cisco Introduction to Cybersecurity & Cybersecurity Essentials — free certificates (netacad.com)

Note: CompTIA Security+ and OSCP are industry-valued but paid exams (~\$400+); study free via Professor Messer/TryHackMe first, then pay only for the exam when ready.

Phase 6 — Build Portfolio & Get Job-Ready (Weeks 37–44)

Topics to cover

- Document your TryHackMe/HTB writeups on a free GitHub Pages blog
- Build 2–3 home-lab projects (e.g., set up a SIEM, detect an attack, write an incident report)

- Contribute to open-source security tools on GitHub
- Practice interview questions & build a resume tailored to SOC Analyst / Security Analyst roles

Free resources

Portfolio & Job Prep

- GitHub Pages — free hosting for your writeup blog (pages.github.com)
- [r/cybersecurity](#) and [r/AskNetsec](#) — free community advice on Reddit
- Cybersecurity job simulations — [Forage.com](#) (free virtual job simulations from real companies)

Suggested weekly rhythm: 40% theory/videos, 40% hands-on labs, 20% writing notes/writeups. Consistency beats intensity — steady weekly progress will get you job-ready faster than binge study.